Information Accessibility and Employability in Non-primary Sector in Burkina Faso

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Capstone Project Final Report

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Abstract

Burkina Faso's economy remains predominantly agricultural, with limited digital infrastructure affecting information access across the country. While information accessibility is theoretically related to labour market outcomes, few studies exist on this relationship in low-income contexts. This study investigates whether and how information accessibility affects employment in non-primary sectors in Burkina Faso. Using the 2018 Harmonized Survey on Household Living Standards data, we employ logistic regression models with instrumental variable approaches to examine causality between information access and non-agricultural job participation. Our analysis highlights that information accessibility significantly increases non-primary sector employment probability by 10.7 percentage points, with additional endogeneity tests confirming this causal relationship. Robustness checks reveal differential impacts across technologies (mobile phones: 13.4pp; internet: 18.8pp) and household characteristics. These findings highlight the importance of expanding digital connectivity as a policy for economic transformation in developing countries and contribute to understanding the mechanisms through which information access impacts labour market outcomes.

Keywords: Information accessibility, employment, non-primary sector, digital adaptation, economic development

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1 Introduction

This study investigates the relationship between information accessibility and employability in non-primary sectors in Burkina Faso in 2018, a developing West African country where digital infrastructures remains limited. Our analysis focuses on how digital information accessibility influences employment in non-agricultural sector, controlling for various demographic and socioeconomic factors.

Burkina Faso's economic landscape is characterised by an agricultural predominance, with approximately 70% of the labour force engaged in primary sector (World Bank, 2019). Arguably, such substantial dependence on agriculture constrains economic diversification and sustainable development in the region. Previous assessment also highlights a stark urban-rural digital divide, limiting employment opportunities for the majority of populations, particularly in remote regions (Imran et al., 2014).

Our research is grounded in several economic theories: Human Capital Theory (Becker, 1962) suggests better information access can enable essential skill acquisition and career opportunities. In addition, Information Economics (Stiglitz, 2000) shows that digital connectivity minimises information asymmetry and enhance labour market efficiency. Structural Transformation Theory (McMillan & Rodrik, 2011) highlights how technological adaptation and information access are crucial for workforce participation as economies transition from agricultural to manufacturing and service sectors.

Therefore, understanding the causal relationship between digital information access and labour market outcomes in Burkina Faso provides valuable insights for economic development and target infrastructure investments.

2 Data[1]

2.1 Variable Definitions [2]

2.1.1 Outcome Variable

Employment Status (job_status): A binary variable equaling 1 if an individual has worked in the non-agricultural sector for at least one hour in the past week. This includes self-employment, government employment, and trainee positions, aligning with standards for labour force participation measurement (International Labour Organisation, 2024).

2.1.2 Variable of Interest

Information Accessibility (accessinfo): A binary variable equaling 1 if an individual has access to at least one of the following technologies: internet, mobile phone, telephone, or television. This composite measure captures various channels through which individuals may access information relevant to employment opportunities and market conditions.

2.1.3 Control Variables

To address potential confounding factors and minimise omitted variable bias, we incorporate 15 control variables across two main categories. See following table for specification and rationale:

[1] Our analysis utilises the 2018-2019 Burkina Faso Harmonised Survey on Household Living Standards (EHCVM), a survey providing detailed information on living conditions, employment status, technology access, and demographic characteristics.

[2] See Tablle A1 for descriptive statistics

Table 1: Controlled Variables

Variable	Description	Rationale
Demographic Characteristics		
female	Equals 1 if female, 0 otherwise	Accounts for gender differences in labour market participation due to social norms and discrimination
age	Age of the individual in years	Controls for life-cycle employment patterns as younger and older individuals typically have lower participation rates
age2	Square of the individual's age	Captures non-linear relationship between age and employment, accounting for declining participation rates at older ages
ifdiploma	Equals 1 if received abt diploma or equivalent, 0 otherwise	Controls for human capital differences affecting employment opportunities and wage potential
ifmarried	Equals 1 if the individual is married, 0 otherwise	Accounts for family structure impact on labour market decisions
liv_for_6mth	Equals 1 if the individual has lived in the location for at least 6 months, 0 otherwise	Controls for residential stability that may affect job search and employment
Socioeconomic Factors		
ifrural	Equals 1 if the individual lives in rural area, 0 otherwise	Controls for urban-rural differences in job opportunities and infrastructure
ifroad	Equals 1 if the individual has access to roads, 0 otherwise	Accounts for physical connectivity to markets and employment centers
localdist	Distance to local town/village center	Indicates geographic barriers from economic centers
ifbus	Equals 1 if individual has access to bus services available, 0 otherwise	Accounts for transportation infrastructure affecting physical job access
train_cen	Equals 1 if vocational training center exists locally, 0 otherwise	Controls for local skill development infrastructure independent of information access
Enterprises	Equals 1 if the individual's household owns non-agricultural enterprises, 0 otherwise	Controls for individual's local economic development and job availability
hhsize	Number of people in household	Accounts for family responsibilities and labour supply decisions
lnminmonthcost	Natural log of minimum monthly cost	A proxy for household wealth measurement, affecting technology adoption and employment
rent	Monthly rent payment (in thousands)	Additional measure of household economic status

3 Econometric Model and Methods

Using linear probability model, the baseline model specification follows:

$$\mathbb{E}[Y_i] = \mathbb{P}[Y_i = 1|X_i'\beta] = X_i'\beta \tag{1}$$

Where Y_i denotes employment status, X_i denotes our covariates.

Following above model, we estimate two model specifications:

1. **Simple regression:** Employment on information access only

$$\mathbb{P}[\text{job_status}_i = 1|\text{accessinfo}_i] = \beta_0 + \beta_1 \cdot \text{accessinfo}_i \tag{2}$$

2. **Multiple regression:** Adding demographic and socioeconomic controls

$$\mathbb{P}[\text{job_status}_i = 1|\text{accessinfo}_i, D_j, S_k] = \alpha_0 + \alpha_1 \cdot \text{accessinfo}_i + \sum_{j=1}^{m} \delta_j \cdot D_j + \sum_{k=1}^{n} \theta_k \cdot S_k \tag{3}$$

   Where $\mathbf{D_j}$ denotes demographic characteristics, and $\mathbf{S_k}$ denotes socioeconomic factors

4 Regression Results and Interpretation

4.1 Baseline Results[3]

Table 2: Key regression results

	Dependent variable:	
	job_status	
	Simple	Multiple
accessinfo	0.270***	0.107***
	(0.005)	(0.005)
Full Controls	No	Yes
Observations	27,019	27,019
R^2	0.092	0.233
F Statistic	2,724.094***	511.364***

4.1.1 Simple vs. Multiple Regression Analysis

Simple Regression Model (0.270*)**: Having access to information, individuals are 27% more likely to have a job in non-agricultural sector. However, this result should be treated with caution as this estimator is likely to reflect both the direct and indirect effects of information accessibility. **Multiple Regression Model (0.107***)**: After controlling for demographic and socioeconomic factors, the likelihood of non-agricultural employment reduced to 10.7%. This reduction highlights a substantial variable bias that has been omitted in the simple model. However, consistent statistical significance suggests a potentially causal relationship.

4.2 Controlled Variable Effects

The 16.3% difference in marginal effects underscores several critical implications:

1. **Confounding effects:** Omitting variables such as entrepreneurship and wealth incorrectly impose their effect on information accessibility and employment outcomes simultaneously.

2. **Socioeconomic Stratification:** Access to information is arguably correlated with a higher socioeconomic status, such as rent, which independently affects employment.

3. **Demographic Determinants**: Gender and household size can significantly influence the outcome of the labor market.

4. **Geographic Disparities:** Residence in urban or rural areas could affect the quality of access to information and employment opportunities.

[3]See Table A2 for full results

5 Endogeneity Analysis

5.1 Potential Simultaneity Bias

Intuitively, having limited information accessibility constrains job-finding efficiency, and being unemployed further impedes information attainment. This would lead the key independent variable (**accessinfo**) to become biased and inconsistent.

5.2 Instrumental Variable Candidates

Using instrumental variables (IV) approach, we employ two sets of instruments to address potential endogeneity:

- **Electricity bill (elecbill):** Monthly electricity expenditure. This is a valid instrument because:
  - **Relevance:** Electricity availability and consumption are necessary for operating any of the information technologies, making it strongly relevant to information access.
  - **Exogeneity:** Electricity expenses are primarily determined by infrastructure availability and household capacity, thereby unlikely to be related to employment status in non-primary sectors, thereby satisfying the exogeneity condition. Although electricity may correlate with factors like urban residence, it should still be exogenous after controlling for these factors[4].
- **Mobile Service Quality:** A set of 3 binary variables indicating the quality of mobile service connections from 3 main providers.
  - **Relevance:** Mobile service provider quality directly impacts information accessibility as it determines connection reliability and data speeds.
  - **Exogeneity:** Provider coverage are generally related on technical and geographical factors rather than individual employment characteristics, making them exogenous to our dependent variable. Again, the potential correlation between these IVs and urban residences can be controlled with such factors[5].

F-stat Results: The joint F-statistic of these IVs is 42.024, which shows statistical validity[6].

[4]See Table 1
[5]See Table 1
[6]See Table A3 for full results

5.3 IV Implementation[7]

Table 3: Key IV Regression Result

	Dependent variable: job_status	
	OLS	IV
accessinfo	0.107***	0.051
	(0.005)	(0.081)
Full Controls	Yes	Yes
Observations	27,019	27,019
Adjusted R^2	0.232	0.229
F Statistic	511.364***	
Note: *p<0.1; **p<0.05; ***p<0.01		

In Table 3. Interestingly, the slope coefficient of **accessinfo** reduces and becomes insignificant with a relatively large standard error under IV estimation. This necessitates further evaluation on the diagnostic test results.

Table 4: Diagnostic Tests

	df1	df2	statistic	p-value
Weak instruments	4	26999	42.024	$< 2e-16^{***}$
Wu-Hausman	1	27001	0.639	0.4241
Sargan	3	NA	9.476	0.0236*

In Table 4, while strong instrumental relevance remains, the p-value of Wu-Hausman test (0.4241) suggests that **accessinfo** might not be endogeneous. More importantly, the p-value of Sargan test (0.0236) for overidentification reveals that at least one instrument variable is endogeneous. Further investigation on instruments validity is required.

[7]See Table A5 for full results

5.4 IV Reselection Through Overidentification Test

5.4.1 Overidentification Test

Table 5: Overidentification Test Results

Hypothesis	Chi Square	P Value
elecbill=0, phone1=0, phone2=0, phone3=0	11.205	0.024
phone1=0, phone2=0, phone3=0	11.172	0.011
elecbill=0, phone1=0, phone2=0	9.337	0.025
elecbill=0, phone1=0, phone3=0	7.088	0.069
elecbill=0, phone2=0, phone3=0	10.178	0.017
elecbill=0, phone1=0	**0.604**	**0.740**
elecbill=0, phone2=0	6.119	0.047
elecbill=0, phone3=0	7.088	0.029
phone1=0, phone2=0	9.306	0.010
phone1=0, phone3=0	7.064	0.029
phone2=0, phone3=0	10.147	0.006

From Table 5, after conducting a sequence of overidentification tests, only the combination of **elecbill** and **phone1** maintains the null hypothesis; no endogeneity in instruments, thereby making it the most appropriate.

5.4.2 Implementation of Updated Instruments [8]

Given the IV estimation with IVs (**elecbill** & **phone1**) that are free of endogeneity:

Table 6: IV Regression Results and Hausman Test

	Dependent variable: job_status		
	OLS	IV	New IV
	(1)	(2)	(3)
accessinfo	0.107***	0.051	0.077
	(0.005)	(0.081)	(0.089)
Full Controls	Yes	Yes	Yes
Observations	27,019	27,019	27,019
Adjusted R^2	0.232	0.229	0.231

Note: Robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

[8] See Table A6 for full result

Table 7: Diagnostic Tests

	df1	df2	statistic	p-value
Weak instruments	2	27001	73.379	$< 2e-16^{***}$
Wu-Hausman	1	27001	0.156	0.693
Sargan	1	NA	0.539	0.463

While both the weak instrument and Sargan test verifies verifies instrumental validity of our new IV set, the slope coefficient [9] of **accessinfo** is still insignificant. Meanwhile, the insignificant Hausman test result should implies exogeneity and asymptotic similarity between OLS and IV estimators, which is contradicted by our results. These mixed results entails a more thorough endogeneity analysis.

5.5 Component-Level Endogeneity Testing

5.5.1 Decomposition of Main Independent Variable

To extend our analysis, we decompose **accessinfo** back to the 4 original binary variables and further test for endogeneity on a component-level:

Table 8: Variable Definitions

Variable	**Definition**
ifphone	Equals 1 if the individual has access to smartphone and 0 otherwise.
ifinternet	Equals 1 if the individual has access to internet and 0 otherwise.
iftelephone	Equals 1 if the individual has access to telephone and 0 otherwise.
ifTV	Equals 1 if the individual has TV connection and 0 otherwise.

[9] See Table A3 and Table A4 for full specification

5.5.2 Regression Results[10]

Table 9: Component-Level Analysis: OLS and IV Estimates with Robust Standard Errors

	Dependent variable: Employment Status							
	Mobile Phone		Internet		Television		Landline	
	OLS	IV	OLS	IV	OLS	IV	OLS	IV
ifphone	0.134***	0.143						
	(0.005)	(0.151)						
ifinternet			0.188***	0.045				
			(0.010)	(0.069)				
ifTV					0.022**	0.024		
					(0.010)	(0.032)		
iftelephone							0.013	0.177
							(0.044)	(0.267)
Observations	27,019	27,019	27,019	27,019	27,019	27,019	27,019	27,019
Adjusted R^2	0.239	0.239	0.235	0.227	0.222	0.222	0.221	0.221

Notes: Robust standard errors in parentheses.
* p<0.1; ** p<0.05; *** p<0.01

All coefficients become statistically insignificant after controlling for endogeneity. This finding follows the trend of our previous IV analysis.

5.5.3 Component Level Diagnostics[11]

Table 10: IV Diagnostic Tests Results

	Information Technology Components			
	Mobile Phone	Internet	Television	Landline
Diagnostic Tests				
Weak Instruments Test	23.79***	310.48***	1597.87***	372.17***
	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
Wu-Hausman Test	0.01	6.29**	0.00	0.58
	(0.943)	(0.012)	(0.951)	(0.448)
Sargan Test	0.39	1.02	0.81	0.93
	(0.530)	(0.314)	(0.370)	(0.334)

Notes: P-values in parentheses.
* p<0.1; ** p<0.05; *** p<0.01

[10] See Table A7 for full results
[11] See Table A8 for Full result

This set of IV diagnostic tests reveals that only internet access exhibits significant endogeneity (Wu-Hausman test = 6.29, p = 0.012). Meanwhile, all instruments are statistically strong and valid, confirming the robustness of the identification strategy. In light of this, we recombine the remaining 3 variables into a new main independent variable, **accessinfo_new**, for further analysis.

5.5.4 New Model Specification[12]

Main OLS and IV implementation results are presented in the table below:

Table 11: New Information Accessibility

	Dependent variable:	
	job_status	
	OLS	IV
accessinfo_new	0.108***	0.077
	(0.005)	(0.089)
Weak Instruments Test		74***
Wu-Hausman Test		0.174
Sargan Test		0.460
Observations	27,019	27,019
Adjusted R^2	0.232	0.231

Note: *p<0.1; **p<0.05; ***p<0.01

Notably, we observe a consistent pattern where the coefficient becomes statistically insignificant under IV estimation in conjunction with diagnostic test failing to detect substantial endogeneity.

This may lead to a counterintuitive conclusion that OLS method is in fact appropriate with limited endogeneity concerns. Nevertheless, there may still be certain limitations to our IV candidates, these will be discussed in later sections.

[12]See Table A9 for full results

6 Robustness Checks

6.1 Alternate Dependent Variables

To examine whether the effect of information accessibility has similar effects on dependent variables of different measures. We estimate OLS again with an alternative economic outcome measure[13]; **Main_Job** which Equals 1 if the individual's primary employment is/was in the non-agricultural sector and 0 otherwise.

Table 12: Original VS. Alternate Dependent variable

	Dependent variable:	
	job_status	main_job
	(1)	(2)
accessinfo_new	0.108***	0.052***
	(0.005)	(0.007)
Observations	27,019	16,168
Adjusted R^2	0.232	0.431
F Statistic	511.907***	766.613***

Note: *p<0.1; **p<0.05; ***p<0.01

From Table 12, despite the marginal effect of **accessinfo_new** on **main_job** is smaller, the direction and statistical significance remain consistent in both specification. This consistency across different dependent variables strengthens our confidence in the main finding, i.e., information access positively correlates non-agricultural employment.

6.2 Subsample Analysis

To further examine whether this casual relationship persists across different subgroups. We divided our sample along two dimensions[14]:

1. **Household Size**: Divide the sample into 2 sub-samples based on whether a household has below 10 members or more

2. **Minimum Monthly Cost**: Divide sample into 2 sub-samples based on whether a household's minimum monthly cost is below 110,000 CFA francs or more

[13]See Table A10 for full result
[14]See Table A11 and Table A12 for descriptive statistics

6.2.1 Regression Results Interpretation[15]

Table 13: Subgroup Analysis: Household Size and Monthly Cost

	Dependent variable:			
	job_status			
	Household Size < 10	Household Size ≥ 10	Min. Monthly Cost < 110,000	Min. Monthly Cost ≥ 110,000
	(1)	(2)	(3)	(4)
Information Access (New)	0.129***	0.065***	0.113***	0.105***
	(0.007)	(0.008)	(0.007)	(0.008)
Observations	18,126	8,893	13,894	13,125
Adjusted R^2	0.254	0.162	0.213	0.247

Note: *p<0.1; **p<0.05; ***p<0.01

As shown, information accessibility maintains a positive and statistically significant relationship with employment status across all subgroups, which confirms the robustness of our main result. This consistency is particularly noteworthy given the differences between these subgroups in terms of their demographic and economic characteristics.

6.2.2 Z-test validation

After conducting one-side z-tests for differences in slope coefficients:

Table 14: Z-Test Results:

	Coefficients				P-Values
Alternative-Hypothesis	Group 1	Group 2	Difference	Z-Statistic	One-Sided
Small HH > Large HH	0.1293	0.0652	0.0640	5.8203	0.0000
Low Cost > High Cost	0.1126	0.1048	0.0078	0.6972	0.2428

Notes:
Group 1: Small HH (≤10 members), Low Cost (≤110,000)
Group 2: Large HH (>10 members), High Cost (>110,000)

Information accessibility has statistically larger effect on non-agricultural employment for small households ($\beta = 0.1293$) compared to large households ($\beta = 0.0652$). In contrast, the monthly cost analysis reveals no significant heterogeneity, i.e., the effect of information accessibility is consistent throughout all income levels.

[15]See Table A13 for full results

7 Extensions

7.1 Models and Specifications

While our analysis regarding LPM has been robust, we acknowledge several limitations to this specification:

- $\mathbf{x_i}\beta$ can be outside [0,1].
- Heteroscedastic errors depends on $\mathbf{X_i}$.
- Non-normal errors for binary outcomes.

To address above limitations and ensure a valid statistical inference, we extend our research by adopting a logit model to examine the same set of variables (Xi, et al. 2019):

The baseline model specification follows:

$$\mathbb{E}[Y_i = 1|X_i'\beta] = G(\boldsymbol{X_i'\beta}) = \frac{e^{X_i'\beta}}{1 + e^{X_i'\beta}} \tag{4}$$

Where Y represents employment status, X denotes our covariates, and $G(\cdot)$ is the logistic cumulative distribution function.

Following above model, our specification becomes:

$$\mathbb{P}[\text{job_status}_i = 1|accessinfo_i, D_j, S_k] = G\left(\alpha_0 + \alpha_1 \cdot accessinfo_i + \sum_{j=1}^{m} \delta_j D_j + \sum_{k=1}^{n} \theta_k S_k\right) \tag{5}$$

Where D_{ji} represents demographic characteristics, and S_{ki} represents socioeconomic factors.

7.2 Results Interpretation[16]

Table 15: Model Comparison: Linear Probability vs Logit Estimation Results

Variable	Linear Probability Model	Logit Model	
		Coefficient	Average Marginal Effect
Access to Information	0.107***	0.945***	0.132***
	(0.005)	(0.043)	(0.006)
Model Diagnostics			
F-statistic	511.364***	-	-
Log Likelihood	-	-11,669.770	-11,669.770
AIC	-	23,373.540	23,373.540
Model Specifications			
Full Controls	Yes	Yes	Yes

*** p<0.001, ** p<0.01, * p<0.05

Given the coefficient of a logit model shows the log-odd, we convert the coefficients to an average marginal effect to facilitate interpretation. The AME indicates individuals with access to information has a 13.2% higher probability of non-agriculture employment. This is consistent with our results obtained from the LPM model, providing reassurance about the robustness of our core result across different estimation approaches.

[16]See Table A15 for full result

8 Conclusion

Our analysis reveals that **information accessibility significantly increases the probability of non-agricultural employment by 10.7%** after controlling for demographic and socioeconomic factors. This represents a substantial effect size, particularly given that only 25.4% of individuals in our sample are employed in non-primary sectors.

Our extensive endogeneity analysis using instrumental variables suggests **concerns about simultaneity bias may be limited**. Despite employing multiple identification strategies, IV estimates consistently lose statistical significance while maintaining similar magnitudes to OLS results, suggesting our OLS estimates likely capture genuine causal relationships rather than spurious correlations.

Additional limitations include the cross-sectional nature limiting long-term causal inference, and unquantifiable factors such as digital literacy, social networks, and terrorist activities in Burkina Faso during 2018 potentially distorting labour market dynamics (United States Department of State,2019).

More importantly, our endogeneity analysis should be treated with caution. Our IV may still be suffering from certain endogeneity concerns: electricity consumption may correlate with employment status where individuals working remotely can incur higher bills. In addition, selection bias exists where households still with formal electricity connections and receive bills during such social disturbance are likely to be wealthier, more urban, or have better infrastructure access than those without formal connections.

This study provides evidence for a positive relationship between information accessibility and non-agricultural employment in Burkina Faso. Future research should employ longitudinal data to capture dynamic effects and explore underlying mechanisms affecting employment outcomes.

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Appendix 1: Tables and Figures

Table A1: Descriptive Statistics

	N	Mean	St. Dev.	Min	Max
Main variables					
accessinfo	27,019	0.611	0.488	0	1
job_status	27,019	0.254	0.435	0	1
Controlled Variables					
female	27,019	0.537	0.499	0	1
min_month_cost	27,019	178,173.500	251,536.600	70	5,000,000
ifrural	27,019	0.569	0.495	0	1
age	27,019	27.010	20.511	0	100
ifroad	27,019	0.297	0.457	0	1
train_cen	27,019	0.160	0.367	0	1
ifbus	27,019	0.157	0.363	0	1
ifmarried	27,019	0.540	0.498	0	1
ifdiploma	27,019	0.187	0.390	0	1
hhsize	27,019	8.884	5.763	1	51
ifTV	27,019	0.106	0.307	0	1
ifinternet	27,019	0.101	0.301	0	1
ifphone	27,019	0.589	0.492	0	1
iftelephone	27,019	0.004	0.066	0	1
localdist	27,019	19.982	25.304	0	145
liv_for_6mth	27,019	0.448	0.497	0	1
rent	27,019	1.405	6.968	0.000	350.000
Enterprises	27,019	0.559	0.497	0	1
IV Candidates					
elecbill	27,019	2.438	7.058	0.000	170.000
phone1	27,019	0.807	0.395	0	1
phone2	27,019	0.829	0.376	0	1
phone3	27,019	0.662	0.473	0	1

Table A2: Simple and Multiple LPM

	Dependent variable:	
	job_status	
	Simple Regression	Multiple Regression
accessinfo	0.270***	0.107***
	(0.005)	(0.005)
ifrural		-0.088***
		(0.008)
ifroad		0.054***
		(0.007)
localdist		-0.0003***
		(0.0001)
liv_for_6mth		0.081***
		(0.007)
train_cen		-0.013
		(0.008)
ifbus		-0.001
		(0.008)
ifmarried		0.168***
		(0.005)
female		-0.066***
		(0.005)
age		0.001***
		(0.0004)
age2		-0.00002***
		(0.00001)
lnminmonthcost		0.025***
		(0.003)
ifdiploma		0.007
		(0.007)
rent		0.003***
		(0.0005)
hhsize		-0.009***
		(0.0004)
Enterprises		0.170***
		(0.005)
Constant	0.089***	-0.180***
	(0.003)	(0.034)
Observations	27,019	27,019
R^2	0.092	0.233
Adjusted R^2	0.092	0.232
Residual Std. Error	0.415	0.382
F Statistic	2,724.094***	511.364***

Note: *p<0.1; **p<0.05; ***p<0.01

Table A3: F-test for Instrument Relevance (4 IVs)

Model 1: restricted model

Model 2: accessinfo ∼ elecbill + phone1 + phone2 + phone3 + ifrural + ifroad + localdist + liv_for_6mth + train_cen + ifbus + ifmarried + female + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize + Enterprises

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	27003	4810.2				
2	26999	4780.4	4	29.763	42.024	< 2.2e-16***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table A4: F-test for Instrument Relevance (2 IVs)

Model 1: restricted model

Model 2: accessinfo ∼ elecbill + phone1 + ifrural + ifroad + localdist + liv_for_6mth + train_cen + ifbus + ifmarried + female + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize + Enterprises

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	27003	4810.2				
2	27001	4784.2	2	26.003	73.379	< 2.2e-16***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table A5: Multiple and IV Regression

	Dependent variable: job_status	
	OLS	*IV*
accessinfo	0.107***	0.051
	(0.005)	(0.081)
ifrural	-0.088***	-0.094***
	(0.008)	(0.013)
ifroad	0.054***	0.059***
	(0.007)	(0.010)
localdist	-0.0003***	-0.0003***
	(0.0001)	(0.0001)
liv_for_6mth	0.081***	0.086***
	(0.007)	(0.010)
train_cen	-0.013	-0.012
	(0.008)	(0.008)
ifbus	-0.001	-0.001
	(0.008)	(0.008)
ifmarried	0.168***	0.186***
	(0.005)	(0.027)
female	-0.066***	-0.077***
	(0.005)	(0.016)
age	0.001***	0.001***
	(0.0004)	(0.0004)
age2	-0.00002***	-0.00002***
	(0.00001)	(0.00001)
lnminmonthcost	0.025***	0.028***
	(0.003)	(0.005)
ifdiploma	0.007	0.013
	(0.007)	(0.011)
rent	0.003***	0.003***
	(0.0005)	(0.001)
hhsize	-0.009***	-0.009***
	(0.0004)	(0.001)
Enterprises	0.170***	0.174***
	(0.005)	(0.007)
Constant	-0.180***	-0.186***
	(0.034)	(0.035)
Observations	27,019	27,019
R^2	0.233	0.230
Adjusted R^2	0.232	0.229
Residual Std. Error	0.382	0.382
F Statistic	511.364***	

Note: *p<0.1; **p<0.05; ***p<0.01

Table A6: IV and Hausman Test Results

	Dependent variable:	
	job_status	
	New IV	*Hauseman Test*
accessinfo	0.077	0.077
	(0.089)	(0.089)
residuals		0.030
		(0.089)
ifrural	−0.091***	−0.091***
	(0.013)	(0.013)
ifroad	0.057***	0.057***
	(0.010)	(0.010)
localdist	−0.0003***	−0.0003***
	(0.0001)	(0.0001)
liv_for_6mth	0.084***	0.084***
	(0.010)	(0.010)
train_cen	−0.013	−0.013
	(0.008)	(0.008)
ifbus	−0.001	−0.001
	(0.008)	(0.008)
ifmarried	0.177***	0.177***
	(0.030)	(0.030)
female	−0.072***	−0.072***
	(0.018)	(0.018)
age	0.001***	0.001***
	(0.0004)	(0.0004)
age2	−0.00002***	−0.00002***
	(0.00001)	(0.00001)
lnminmonthcost	0.027***	0.027***
	(0.005)	(0.005)
ifdiploma	0.010	0.010
	(0.012)	(0.012)
rent	0.003***	0.003***
	(0.001)	(0.001)
hhsize	−0.009***	−0.009***
	(0.001)	(0.001)
Enterprises	0.172***	0.172***
	(0.007)	(0.007)
Constant	−0.183***	−0.183***
	(0.035)	(0.035)
Observations	27,019	27,019
R^2	0.232	0.233
Adjusted R^2	0.231	0.232
Residual Std. Error	0.382	0.382
F Statistic		481.278***

Note: *p<0.1; **p<0.05; ***p<0.01

Table A7: Component-Level Analysis: OLS and IV Estimates with Robust Standard Errors

	Dependent variable: Employment Status							
	Mobile Phone		Internet		Television		Landline	
	OLS	IV	OLS	IV	OLS	IV	OLS	IV
Mobile Phone Access	0.134***	0.143						
	(0.005)	(0.151)						
Internet Access			0.188***	0.045				
			(0.010)	(0.069)				
TV Access					0.022**	0.024		
					(0.010)	(0.032)		
Landline Access							0.013	0.177
							(0.044)	(0.267)
Full Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	27,019	27,019	27,019	27,019	27,019	27,019	27,019	27,019
R^2	0.239	0.239	0.235	0.227	0.222	0.222	0.222	0.221
Adjusted R^2	0.239	0.239	0.235	0.227	0.222	0.222	0.221	0.221
Residual Std. Error	0.380	0.380	0.381	0.383	0.384	0.384	0.384	0.384

Notes: Robust standard errors in parentheses.
All regressions include the full set of control variables as specified in the methodology section.
Instrumental variables for IV regressions are electricity bill (elecbill) and availability of phone services in the community (phone1).
* p<0.1; ** p<0.05; *** p<0.01

Table A8: IV Diagnostic Tests Results

	Information Accessibility Components			
	Mobile Phone	Internet	Television	Landline
Diagnostic Tests				
Weak Instruments Test	23.79***	310.48***	1597.87***	372.17***
	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)
Wu-Hausman Test	0.01	6.29**	0.00	0.58
	(0.943)	(0.012)	(0.951)	(0.448)
Sargan Test	0.39	1.02	0.81	0.93
	(0.530)	(0.314)	(0.370)	(0.334)
Model Info				
Instruments	elecbill, phone1	elecbill, phone1	elecbill, phone1	elecbill, phone1
Full Controls	Yes	Yes	Yes	Yes
Observations	27,019	27,019	27,019	27,019
R^2	0.239	0.227	0.222	0.221
Adjusted R^2	0.239	0.227	0.222	0.221

Note: P-values of each test result in parentheses
Instruments: elecbill (electricity bill) and phone1 (phone ownership).
Weak instruments test: H0 = instruments are weak.
Wu-Hausman test: H0 = no endogeneity (OLS consistent).
Sargan test: H0 = instruments are valid (exogenous).

Table A9: New Information Accessibility

	Dependent variable:	
	job_status	
	OLS	IV
accessinfo_new	0.108***	0.077
	(0.005)	(0.089)
ifrural	-0.088***	-0.091***
	(0.008)	(0.013)
ifroad	0.054***	0.057***
	(0.007)	(0.010)
localdist	-0.0003***	-0.0003***
	(0.0001)	(0.0001)
liv_for_6mth	0.081***	0.084***
	(0.007)	(0.010)
train_cen	-0.013	-0.013
	(0.008)	(0.008)
ifbus	-0.001	-0.001
	(0.008)	(0.008)
ifmarried	0.167***	0.177***
	(0.005)	(0.030)
female	-0.066***	-0.072***
	(0.005)	(0.017)
age	0.001***	0.001***
	(0.0004)	(0.0004)
age2	-0.00002***	-0.00002***
	(0.00001)	(0.00001)
lnminmonthcost	0.025***	0.027***
	(0.003)	(0.005)
ifdiploma	0.007	0.010
	(0.007)	(0.012)
rent	0.003***	0.003***
	(0.0005)	(0.001)
hhsize	-0.009***	-0.009***
	(0.0004)	(0.001)
Enterprises	0.170***	0.172***
	(0.005)	(0.007)
Constant	-0.180***	-0.183***
	(0.034)	(0.035)
Weak Instruments Test		74***
Wu-Hausman Test		0.174
Sargan Test		0.460
Observations	27,019	27,019
R^2	0.233	0.232
Adjusted R^2	0.232	0.231
Residual Std. Error	0.382	0.382
F Statistic	511.907***	

Note: *p<0.1; **p<0.05; ***p<0.01

Table A10: Original VS. Alternate Dependent variable

	Dependent variable:	
	job_status	main_job
	(1)	(2)
accessinfo_new	0.108***	0.052***
	(0.005)	(0.007)
ifrural	-0.088***	-0.190***
	(0.008)	(0.011)
ifroad	0.054***	0.115***
	(0.007)	(0.010)
localdist	-0.0003***	-0.001***
	(0.0001)	(0.0001)
liv_for_6mth	0.081***	0.175***
	(0.007)	(0.010)
train_cen	-0.013	0.028***
	(0.008)	(0.011)
ifbus	-0.001	-0.002
	(0.008)	(0.010)
ifmarried	0.167***	-0.017***
	(0.005)	(0.006)
female	-0.066***	0.011*
	(0.005)	(0.006)
age	0.001***	0.012***
	(0.0004)	(0.001)
age2	-0.00002***	-0.0001***
	(0.00001)	(0.00001)
lnminmonthcost	0.025***	0.072***
	(0.003)	(0.004)
ifdiploma	0.007	0.042***
	(0.007)	(0.008)
rent	0.003***	0.006***
	(0.0005)	(0.001)
hhsize	-0.009***	-0.011***
	(0.0004)	(0.0005)
Enterprises	0.170***	0.116***
	(0.005)	(0.006)
Constant	-0.180***	-0.682***
	(0.034)	(0.044)
Observations	27,019	16,168
R^2	0.233	0.432
Adjusted R^2	0.232	0.431
Residual Std. Error	0.382	0.358
F Statistic	511.907***	766.613***

Note: *p<0.1; **p<0.05; ***p<0.01

Table A11: Descriptive Statistics by Household Size

	Small Household Size (< 10)					Large Household Size (≥ 10)				
Variable	N	Mean	St. Dev.	Min	Max	N	Mean	St. Dev.	Min	Max
female	18,126	0.528	0.499	0	1	8,893	0.556	0.497	0	1
accessinfo	18,126	0.644	0.479	0	1	8,893	0.544	0.498	0	1
job_status	18,126	0.296	0.456	0	1	8,893	0.169	0.375	0	1
min_month_cost	18,126	164,022	234,480	70	5,000,000	8,893	207,017	280,954	5,000	4,000,000
ifrural	18,126	0.507	0.500	0	1	8,893	0.695	0.461	0	1
age	18,126	27.637	20.357	0	100	8,893	25.733	20.764	0	93
ifroad	18,126	0.343	0.475	0	1	8,893	0.202	0.402	0	1
train_cen	18,126	0.178	0.383	0	1	8,893	0.123	0.328	0	1
ifbus	18,126	0.177	0.382	0	1	8,893	0.115	0.320	0	1
ifmarried	18,126	0.549	0.498	0	1	8,893	0.522	0.500	0	1
ifdiploma	18,126	0.222	0.415	0	1	8,893	0.117	0.321	0	1
hhsize	18,126	5.875	2.002	1	9	8,893	15.018	6.056	10	51
ifTV	18,126	0.118	0.323	0	1	8,893	0.080	0.271	0	1
ifinternet	18,126	0.128	0.334	0	1	8,893	0.047	0.211	0	1
ifphone	18,126	0.621	0.485	0	1	8,893	0.523	0.499	0	1
iftelephone	18,126	0.006	0.074	0	1	8,893	0.002	0.044	0	1
localdist	18,126	17.634	23.579	0	145	8,893	24.768	27.895	0	145
liv_for_6mth	18,126	0.502	0.500	0	1	8,893	0.338	0.473	0	1
elecbill	18,126	2.490	7.238	0.000	170.000	8,893	2.332	6.674	0.000	63.500
rent	18,126	2.020	8.342	0.000	350.000	8,893	0.153	1.827	0.000	45.000
Enterprises	18,126	0.534	0.499	0	1	8,893	0.608	0.488	0	1
accessinfo_new	18,126	0.643	0.479	0	1	8,893	0.544	0.498	0	1

Table A12: Descriptive Statistics by Monthly Cost

	Low Cost ($< 110,000$)					High Cost ($\geq 110,000$)				
Variable	N	Mean	St. Dev.	Min	Max	N	Mean	St. Dev.	Min	Max
female	13,894	0.542	0.498	0	1	13,125	0.532	0.499	0	1
accessinfo	13,894	0.544	0.498	0	1	13,125	0.682	0.466	0	1
job_status	13,894	0.213	0.410	0	1	13,125	0.298	0.457	0	1
min_month_cost	13,894	69,087	27,765	70	100,000	13,125	293,652	321,719	110,000	5,000,000
ifrural	13,894	0.658	0.474	0	1	13,125	0.475	0.499	0	1
age	13,894	27.242	20.945	0	100	13,125	26.766	20.038	0	95
ifroad	13,894	0.232	0.422	0	1	13,125	0.366	0.482	0	1
train_cen	13,894	0.138	0.345	0	1	13,125	0.184	0.387	0	1
ifbus	13,894	0.130	0.336	0	1	13,125	0.185	0.389	0	1
ifmarried	13,894	0.548	0.498	0	1	13,125	0.532	0.499	0	1
ifdiploma	13,894	0.141	0.348	0	1	13,125	0.236	0.425	0	1
hhsize	13,894	7.871	4.418	1	33	13,125	9.957	6.744	1	51
ifTV	13,894	0.030	0.170	0	1	13,125	0.186	0.389	0	1
ifinternet	13,894	0.048	0.213	0	1	13,125	0.157	0.364	0	1
ifphone	13,894	0.537	0.499	0	1	13,125	0.643	0.479	0	1
iftelephone	13,894	0.001	0.022	0	1	13,125	0.008	0.091	0	1
localdist	13,894	21.702	24.689	0	145	13,125	18.161	25.817	0	145
liv_for_6mth	13,894	0.358	0.480	0	1	13,125	0.542	0.498	0	1
elecbill	13,894	0.845	3.062	0.000	50.000	13,125	4.124	9.332	0.000	170.000
rent	13,894	0.529	3.041	0.000	75.000	13,125	2.333	9.407	0.000	350.000
Enterprises	13,894	0.520	0.500	0	1	13,125	0.600	0.490	0	1
accessinfo_new	13,894	0.544	0.498	0	1	13,125	0.681	0.466	0	1

Table A13: Subgroup Analysis: Household Size and Monthly Cost

	Dependent variable:			
	job_status			
	Household Size < 10	Household Size ≥ 10	Min. Monthly Cost < 110,000	Min. Monthly Cost ≥ 110,000
	(1)	(2)	(3)	(4)
Information Access (New)	0.129***	0.065***	0.113***	0.105***
	(0.007)	(0.008)	(0.007)	(0.008)
Rural	-0.095***	-0.060***	-0.057***	-0.123***
	(0.010)	(0.014)	(0.010)	(0.012)
Road Access	0.041***	0.066***	0.057***	0.048***
	(0.008)	(0.013)	(0.010)	(0.010)
Local Distance	-0.0003**	-0.0003***	-0.0004***	-0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Living 6+ Months	0.089***	0.063***	0.083***	0.071***
	(0.009)	(0.011)	(0.009)	(0.011)
Training Center	-0.003	-0.030*	-0.026**	-0.001
	(0.010)	(0.016)	(0.012)	(0.011)
Bus Access	0.004	-0.006	0.018	-0.020*
	(0.009)	(0.016)	(0.012)	(0.011)
Married	0.181***	0.124***	0.116***	0.221***
	(0.006)	(0.008)	(0.007)	(0.008)
Female	-0.077***	-0.041***	-0.035***	-0.100***
	(0.006)	(0.008)	(0.006)	(0.007)
Age	0.001***	0.0002	0.001**	0.001
	(0.0005)	(0.001)	(0.0005)	(0.001)
Age Squared	-0.00002***	-0.00001	-0.00002***	-0.00002**
	(0.00001)	(0.00001)	(0.00001)	(0.00001)
Log Min Monthly Cost	0.030***	0.021***	0.021***	0.036***
	(0.004)	(0.005)	(0.005)	(0.007)
Diploma	0.005	-0.018	-0.013	0.020**
	(0.008)	(0.013)	(0.010)	(0.010)
Rent	0.002***	0.003	0.011***	0.002***
	(0.0004)	(0.003)	(0.002)	(0.0004)
Household Size	-0.030***	-0.004***	-0.009***	-0.009***
	(0.002)	(0.001)	(0.001)	(0.0005)
Enterprises	0.190***	0.155***	0.182***	0.156***
	(0.006)	(0.007)	(0.006)	(0.007)
Constant	-0.132***	-0.151***	-0.148**	-0.293***
	(0.043)	(0.056)	(0.060)	(0.088)
Observations	18,126	8,893	13,894	13,125
R^2	0.255	0.163	0.214	0.248
Adjusted R^2	0.254	0.162	0.213	0.247
Residual Std. Error	0.394	0.343	0.363	0.397
F Statistic	387.149***	108.245***	235.672***	270.021***

Note: *p<0.1; **p<0.05; ***p<0.01

Table A14: Logit Model Results: Coefficients and Average Marginal Effects

	Coefficients		Average Marginal Effects	
Variable	Estimate	Std. Error	AME	Std. Error
Access to Information	0.945***	(0.043)	0.132***	(0.006)
Rural	-0.540***	(0.054)	-0.076***	(0.008)
Road Access	0.278***	(0.040)	0.039***	(0.006)
Local Distance	-0.002***	(0.001)	-0.0003***	(0.0001)
Lived 6+ Months	0.534***	(0.049)	0.075***	(0.007)
Training Center	-0.053	(0.044)	-0.007	(0.006)
Bus Access	-0.021	(0.045)	-0.003	(0.006)
Married	1.138***	(0.037)	0.159***	(0.005)
Female	-0.425***	(0.033)	-0.059***	(0.005)
Age	0.006**	(0.003)	0.001**	(0.0004)
Age^2	-0.0001***	(0.00004)	-0.00002***	(0.000006)
Log(Min Monthly Cost)	0.171***	(0.021)	0.024***	(0.003)
Diploma	0.062	(0.042)	0.009	(0.006)
Rent	0.014***	(0.002)	0.002***	(0.0003)
Household Size	-0.083***	(0.004)	-0.012***	(0.0006)
Enterprises	1.292***	(0.037)	0.181***	(0.005)
Constant	-4.492***	(0.245)	–	–
Observations		27,019		
Log Likelihood		-11,669.770		
AIC		23,373.540		

Notes: Robust standard errors in parentheses. AME standard errors calculated using the delta method.
* p<0.1, ** p<0.05, *** p<0.01
The dependent variable is employment status (1 = employed, 0 = not employed).

Table A15: Logit Model Results: Coefficients and Average Marginal Effects

	Logit Coefficients	Average Marginal Effects
accessinfo_new	0.6305	0.0883
	(0.4487)	(0.0628)
residual_hat	0.3162	0.0443
	(0.4498)	(0.0630)
ifrural	-0.5774***	-0.0808***
	(0.0759)	(0.0106)
ifroad	0.3046***	0.0426***
	(0.0552)	(0.0077)
localdist	-0.0022**	-0.0003**
	(0.0008)	(0.0001)
liv_for_6mth	0.5604***	0.0784***
	(0.0618)	(0.0086)
train_cen	-0.0484	-0.0068
	(0.0449)	(0.0063)
ifbus	-0.0198	-0.0028
	(0.0446)	(0.0062)
ifmarried	1.2420***	0.1738***
	(0.1525)	(0.0213)
female	-0.4846***	-0.0678***
	(0.0911)	(0.0127)
age	0.0058*	0.0008*
	(0.0027)	(0.0004)
age2	-0.0001**	-0.0000**
	(0.0000)	(0.0000)
lnminmonthcost	0.1892***	0.0265***
	(0.0328)	(0.0046)
ifdiploma	0.0949	0.0133
	(0.0635)	(0.0089)
rent	0.0146***	0.0020***
	(0.0025)	(0.0003)
hhsize	-0.0846***	-0.0118***
	(0.0042)	(0.0006)
Enterprises	1.3110***	0.1834***
	(0.0453)	(0.0060)
Constant	-4.5410***	–
	(0.2553)	
Observations	N	
AIC	23373.54	
BIC	23513.01	

Notes: Standard errors in parentheses.
* p<0.10, ** p<0.05, *** p<0.01
The dependent variable is employment status (1 = employed in non-agricultural sector, 0 = otherwise).
Logit coefficients represent log-odds ratios; Average Marginal Effects show percentage point changes in probability.

Table A16: Correlation Matrix: Instrumental Variables and Information Access

Variable	Electricity Bill (elecbill)	Phone Service (phone1)	Information Access (accessinfo_new)
Electricity Bill (elecbill)	1.0000	0.1183	0.1935***
Phone Service (phone1)	0.1183	1.0000	0.0973***
Information Access (accessinfo_new)	0.1935***	0.0973***	1.0000

Appendix 2: Mathematical Derivation of the Logistic Regression Model

Problem Setup and Motivation

Binary Classification Framework

Consider a binary response variable Y taking values in $\{0, 1\}$ and a vector of predictors $\mathbf{x} = (x_1, x_2, \ldots, x_p)^T$. Our goal is to model the conditional probability:

$$P(Y = 1|\mathbf{x}) = \pi(\mathbf{x}) \tag{1}$$

Problems with Linear Probability Model

A natural first attempt might be to use a linear model:

$$P(Y = 1|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p = \mathbf{x}^T \boldsymbol{\beta} \tag{2}$$

where $\mathbf{x} = (1, x_1, x_2, \ldots, x_p)^T$ includes the intercept term.

Fundamental Problems:

1. **Range constraint violation:** The linear combination $\mathbf{x}^T\boldsymbol{\beta}$ can take any real value, but probabilities must satisfy $0 \leq P(Y = 1|\mathbf{x}) \leq 1$.
2. **Heteroscedasticity:** For a Bernoulli random variable, $\text{Var}(Y|\mathbf{x}) = \pi(\mathbf{x})[1 - \pi(\mathbf{x})]$, which depends on the mean, violating the constant variance assumption.
3. **Non-linearity of effects:** In reality, the effect of a predictor on probability often exhibits diminishing returns near the boundaries (0 and 1).

The Logit Transformation

Odds and Log-Odds

Definition 1 (Odds)**.** For a probability π, the odds are defined as:

$$\text{Odds} = \frac{\pi}{1 - \pi} \tag{3}$$

The odds represent the ratio of the probability of success to the probability of failure.

Key properties:

- Range: $(0, \infty)$
- Odds $= 1$ when $\pi = 0.5$ (equal probability)
- Odds > 1 when $\pi > 0.5$ (success more likely)
- Odds < 1 when $\pi < 0.5$ (failure more likely)

Definition 2 (Logit Function)**.** The logit (log-odds) function is:

$$\text{logit}(\pi) = \log\left(\frac{\pi}{1-\pi}\right) \tag{4}$$

Key Properties of the Logit:

- Domain: $(0, 1)$
- Range: $(-\infty, \infty)$
- Monotonically increasing
- $\text{logit}(0.5) = 0$
- Symmetric around $\pi = 0.5$

The logit transformation solves our range problem: we can now model the log-odds as a linear function of the predictors without violating probability constraints.

The Logistic Regression Model

We model the log-odds as a linear function of the predictors:

$$\text{logit}(\pi(\mathbf{x})) = \log\left(\frac{\pi(\mathbf{x})}{1-\pi(\mathbf{x})}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p = \mathbf{x}^T \boldsymbol{\beta} \tag{5}$$

Deriving the Logistic Function

To find the explicit form of $\pi(\mathbf{x})$, we solve for π from the logit equation:

$$\log\left(\frac{\pi}{1-\pi}\right) = \mathbf{x}^T\boldsymbol{\beta} \tag{6}$$

$$\frac{\pi}{1-\pi} = \exp(\mathbf{x}^T\boldsymbol{\beta}) \tag{7}$$

$$\pi = (1-\pi)\exp(\mathbf{x}^T\boldsymbol{\beta}) \tag{8}$$

$$\pi = \exp(\mathbf{x}^T\boldsymbol{\beta}) - \pi\exp(\mathbf{x}^T\boldsymbol{\beta}) \tag{9}$$

$$\pi + \pi\exp(\mathbf{x}^T\boldsymbol{\beta}) = \exp(\mathbf{x}^T\boldsymbol{\beta}) \tag{10}$$

$$\pi[1+\exp(\mathbf{x}^T\boldsymbol{\beta})] = \exp(\mathbf{x}^T\boldsymbol{\beta}) \tag{11}$$

$$\pi = \frac{\exp(\mathbf{x}^T\boldsymbol{\beta})}{1+\exp(\mathbf{x}^T\boldsymbol{\beta})} \tag{12}$$

Multiplying numerator and denominator by $\exp(-\mathbf{x}^T\boldsymbol{\beta})$:

$$\pi(\mathbf{x}) = \frac{1}{1+\exp(-\mathbf{x}^T\boldsymbol{\beta})} \tag{13}$$

$$P(Y=1|\mathbf{x}) = \frac{1}{1+\exp(-\mathbf{x}^T\boldsymbol{\beta})} \tag{14}$$

This is the logistic function or sigmoid function.

Properties of the Logistic Function

Let $\sigma(z) = \frac{1}{1+e^{-z}}$ denote the logistic function.

[Properties of the Logistic Function]

1. Range: $\sigma(z) \in (0,1)$ for all $z \in \mathbb{R}$
2. Monotonicity: $\sigma(z)$ is strictly increasing
3. Symmetry: $\sigma(-z) = 1 - \sigma(z)$
4. Limits: $\lim_{z\to-\infty}\sigma(z) = 0$ and $\lim_{z\to+\infty}\sigma(z) = 1$
5. Midpoint: $\sigma(0) = 0.5$
6. Derivative: $\sigma'(z) = \sigma(z)[1-\sigma(z)]$

Proof of Derivative Property:

$$\frac{d}{dz}\sigma(z) = \frac{d}{dz}\frac{1}{1+e^{-z}} \tag{15}$$
$$= \frac{0\cdot(1+e^{-z}) - 1\cdot(-e^{-z})}{(1+e^{-z})^2} \tag{16}$$
$$= \frac{e^{-z}}{(1+e^{-z})^2} \tag{17}$$
$$= \frac{1}{1+e^{-z}}\cdot\frac{e^{-z}}{1+e^{-z}} \tag{18}$$
$$= \frac{1}{1+e^{-z}}\cdot\frac{1+e^{-z}-1}{1+e^{-z}} \tag{19}$$
$$= \sigma(z)[1-\sigma(z)] \tag{20}$$

Parameter Interpretation

Intercept (β_0)

When all predictors equal zero ($\mathbf{x} = \mathbf{0}$):

$$\text{logit}(\pi) = \beta_0 \tag{21}$$
$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 \tag{22}$$
$$\frac{\pi}{1-\pi} = e^{\beta_0} \tag{23}$$

Therefore:

- β_0 is the log-odds when all predictors are zero
- e^{β_0} is the odds when all predictors are zero

Slope Coefficients (β_j)

Consider the change in log-odds when x_j increases by one unit, holding all other variables constant:

$$\Delta\text{logit} = \text{logit}(\pi|x_j+1) - \text{logit}(\pi|x_j) \tag{24}$$
$$= [\beta_0 + \beta_1 x_1 + \cdots + \beta_j(x_j+1) + \cdots + \beta_p x_p] \tag{25}$$
$$- [\beta_0 + \beta_1 x_1 + \cdots + \beta_j x_j + \cdots + \beta_p x_p] \tag{26}$$
$$= \beta_j \tag{27}$$

For the odds ratio:

$$\log\left(\frac{\text{Odds}(x_j+1)}{\text{Odds}(x_j)}\right) = \beta_j \tag{28}$$

$$\frac{\text{Odds}(x_j+1)}{\text{Odds}(x_j)} = e^{\beta_j} \tag{29}$$

Therefore:

- β_j is the change in log-odds for a one-unit increase in x_j
- e^{β_j} is the odds ratio for a one-unit increase in x_j

Individual Observations

For observation i with predictors $\mathbf{x}_i$ and outcome $y_i \in \{0, 1\}$, the probability is:

$$P(Y_i = y_i|\mathbf{x}_i) = \pi_i^{y_i}(1-\pi_i)^{1-y_i} \tag{30}$$

where $\pi_i = P(Y_i = 1|\mathbf{x}_i) = \frac{1}{1+\exp(-\mathbf{x}_i^T\boldsymbol{\beta})}$.

This can be verified:

- When $y_i = 1$: $P(Y_i = 1|\mathbf{x}_i) = \pi_i^1(1-\pi_i)^0 = \pi_i$
- When $y_i = 0$: $P(Y_i = 0|\mathbf{x}_i) = \pi_i^0(1-\pi_i)^1 = 1-\pi_i$

Joint Likelihood

Assuming independence across observations, the joint likelihood is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} P(Y_i = y_i|\mathbf{x}_i) = \prod_{i=1}^{n} \pi_i^{y_i}(1-\pi_i)^{1-y_i} \tag{31}$$

Log-Likelihood

Taking the natural logarithm:

$$\ell(\boldsymbol{\beta}) = \log L(\boldsymbol{\beta}) = \sum_{i=1}^{n}[y_i \log \pi_i + (1-y_i)\log(1-\pi_i)] \tag{32}$$

Substituting the logistic function:

$$\pi_i = \frac{1}{1+e^{-\mathbf{x}_i^T\boldsymbol{\beta}}} \tag{33}$$

$$1-\pi_i = \frac{e^{-\mathbf{x}_i^T\boldsymbol{\beta}}}{1+e^{-\mathbf{x}_i^T\boldsymbol{\beta}}} \tag{34}$$

Therefore:

$$\log \pi_i = -\log(1 + e^{-\mathbf{x}_i^T \boldsymbol{\beta}}) \tag{35}$$

$$\log(1 - \pi_i) = -\mathbf{x}_i^T \boldsymbol{\beta} - \log(1 + e^{-\mathbf{x}_i^T \boldsymbol{\beta}}) \tag{36}$$

Substituting into the log-likelihood:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big[y_i(-\log(1 + e^{-\mathbf{x}_i^T \boldsymbol{\beta}})) \tag{37}$$

$$+(1 - y_i)(-\mathbf{x}_i^T \boldsymbol{\beta} - \log(1 + e^{-\mathbf{x}_i^T \boldsymbol{\beta}})) \Big] \tag{38}$$

$$= \sum_{i=1}^{n} \Big[y_i \mathbf{x}_i^T \boldsymbol{\beta} - \log(1 + e^{\mathbf{x}_i^T \boldsymbol{\beta}}) \Big] \tag{39}$$

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big[y_i \mathbf{x}_i^T \boldsymbol{\beta} - \log(1 + e^{\mathbf{x}_i^T \boldsymbol{\beta}}) \Big] \tag{40}$$

Maximum Likelihood Estimation

Score Function

To find the maximum likelihood estimate, we differentiate the log-likelihood with respect to $\boldsymbol{\beta}$:

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^{n} \left[y_i \mathbf{x}_i - \frac{\mathbf{x}_i e^{\mathbf{x}_i^T \boldsymbol{\beta}}}{1 + e^{\mathbf{x}_i^T \boldsymbol{\beta}}} \right] \tag{41}$$

$$= \sum_{i=1}^{n} \mathbf{x}_i \left[y_i - \frac{e^{\mathbf{x}_i^T \boldsymbol{\beta}}}{1 + e^{\mathbf{x}_i^T \boldsymbol{\beta}}} \right] \tag{42}$$

$$= \sum_{i=1}^{n} \mathbf{x}_i (y_i - \pi_i) \tag{43}$$

In matrix form:

$$U(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \boldsymbol{\beta}} = \mathbf{X}^T (\mathbf{y} - \boldsymbol{\pi}) \tag{44}$$

where $\mathbf{X}$ is the $n \times (p+1)$ design matrix, $\mathbf{y} = (y_1, \ldots, y_n)^T$, and $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_n)^T$.

Normal Equations

Setting the score function equal to zero gives the likelihood equations:

$$\mathbf{X}^T (\mathbf{y} - \boldsymbol{\pi}) = \mathbf{0} \tag{45}$$

This system has no closed-form solution and must be solved iteratively using numerical methods such as Newton-Raphson or Fisher scoring.

Iterative Solution

The Newton-Raphson algorithm updates the parameter estimates as:

$$\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} - \mathbf{H}^{-1}(\boldsymbol{\beta}^{(k)})U(\boldsymbol{\beta}^{(k)}) \tag{46}$$

where $\mathbf{H}$ is the Hessian matrix (matrix of second derivatives):

$$\mathbf{H}(\boldsymbol{\beta}) = -\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^T \pi_i(1-\pi_i) = -\mathbf{X}^T\mathbf{W}\mathbf{X} \tag{47}$$

and $\mathbf{W} = \text{diag}(\pi_1(1-\pi_1), \ldots, \pi_n(1-\pi_n))$ is the weight matrix.

Model Interpretation and Prediction

Probability Prediction

For a new observation with predictors $\mathbf{x}_{\text{new}}$, the predicted probability is:

$$\hat{\pi}_{\text{new}} = \frac{1}{1+\exp(-\mathbf{x}_{\text{new}}^T\hat{\boldsymbol{\beta}})} \tag{48}$$

Classification

A common decision rule is to classify as class 1 if $\hat{\pi} > 0.5$:

$$\begin{aligned}
\hat{\pi} > 0.5 &\Leftrightarrow \frac{1}{1+\exp(-\mathbf{x}^T\hat{\boldsymbol{\beta}})} > 0.5 && (49)\\
&\Leftrightarrow 1 > 1+\exp(-\mathbf{x}^T\hat{\boldsymbol{\beta}}) && (50)\\
&\Leftrightarrow 0 > \exp(-\mathbf{x}^T\hat{\boldsymbol{\beta}}) && (51)\\
&\Leftrightarrow 0 > -\mathbf{x}^T\hat{\boldsymbol{\beta}} && (52)\\
&\Leftrightarrow \mathbf{x}^T\hat{\boldsymbol{\beta}} > 0 && (53)
\end{aligned}$$

Therefore, the decision boundary is the hyperplane $\mathbf{x}^T\hat{\boldsymbol{\beta}} = 0$.

Summary

The logistic regression model is constructed through the following key steps:

1. Recognize limitations of the linear probability model for binary outcomes
2. Transform the problem using the logit function to map probabilities to the real line
3. Model the log-odds as a linear function of predictors
4. Derive the logistic function by inverting the logit transformation

5. Construct the likelihood based on the Bernoulli distribution

6. Estimate parameters using maximum likelihood via iterative algorithms

The resulting model naturally handles the constraints of probability modeling while providing interpretable parameters through odds ratios. The S-shaped logistic curve captures the non-linear relationship between predictors and probabilities, with effects diminishing near the boundaries, which is often realistic for binary outcomes.

Appendix 3: R Code

This appendix contains the R code used for all analysis in this paper. The code is organised by analytical sections.

Setup and Data Loading

```
rm(list = ls())
library(stargazer)
library(tidyverse)
library(magrittr)
library(margins)
library(caret)
library(dplyr)
library(glmnet)
library(car)
library(lmtest)
library(AER)
library(ggplot2)
library(mfx)
library(hdm)
library(modelsummary)
library(estimatr)
library(fixest)
library(sandwich)

# Data files loading
s101me <- read.csv("h.enterprises.csv")
s08ame <- read.csv("h.foodsecurity.csv")
s00me <- read.csv("h.household.csv")
s20me <- read.csv("h.poverty.csv")
s02me <- read.csv("i.education.csv")
s04me <- read.csv("i.employment.csv")
s03me <- read.csv("i.general_health.csv")
s01me <- read.csv("i.socio.csv")
s02c0 <- read.csv("co.socialservice.csv")
s01c0 <- read.csv("co.general.csv")
s11me <- read.csv("s11_me_bfa2018.csv")
ind <- read.csv("ehcvm_individu_bfa2018.csv")
hhd <- read.csv("ehcvm_menage_bfa2018.csv")
```

```
welfare <- read.csv("ehcvm_welfare_bfa2018.csv")
```

Data Merging and Cleaning

```
# Merging data files
household_data <- Reduce(function(x, y) merge(x, y, by = c("grappe","menage"),
                                              all.x = TRUE),
                         list(s101me, s08ame, s00me, s20me, s11me, welfare, ind))

individual_data <- Reduce(function(x, y) merge(x, y, by = c("grappe","menage","s01q00a"),
                                               all.x = TRUE),
                          list(s02me, s04me, s03me, s01me))

community_data <- Reduce(function(x, y) merge(x, y, by = "grappe",
                                              all.x = TRUE),
                         list(s01c0,s02c0))

library(janitor)
individual_data <- individual_data %>% clean_names()
household_data <- household_data %>% clean_names()
community_data <- community_data %>% clean_names()

# Remove duplicates
community_data <- community_data %>%
  distinct(grappe, .keep_all = TRUE)

household_data <- household_data %>%
  distinct(grappe, menage, .keep_all = TRUE)

# Final merge
final_df <- individual_data %>%
  left_join(household_data, by = c("grappe", "menage")) %>%
  left_join(community_data, by = "grappe") %>%
  rename_with(~ gsub("\\.x$", "_ind", .x)) %>%
  rename_with(~ gsub("\\.y$", "_hh", .x))
```

Variable Selection and Transformation

```
# Selecting key variables
df1 <- final_df[, c("s01q01_ind", "s01q07_ind", "s02q29","s01q36", "s04q06",
```

```
              "s04q07","s04q08","s04q09", "s01q03c", "s00q04", "s20q06",
              "s01q39_1", "s01q39_2", "s01q39_3", "s01q39_4", "s01q39_5",
              "s02q01_16", "s02q01_25", "s02q01_24", "s02q01_15", "s01q11_hh",
              "s01q13_1", "s01q13_2", "s01q13_3", "s10q11", "s11q02","s11q34",
              "s11q43","s11q50", "diplome","hhsize","age","s01q05_hh","s01q12_hh",
              "s11q37a", "s11q49", "s11q05", "branch")]

# Variable naming for clarity
names(df1) <- c("female", "ifmarried","educ","ifphone", "iffield", "work_self",
                "work_gov", "work_trainee", "dob", "ifrural", "min_month_cost",
                "inter_phone", "inter_office", "inter_cafe", "inter_home",
                "inter_school", "ifroad", "train_cen", "liter_cen", "ifbus",
                "elec_work", "phone1", "phone2", "phone3", "Enterprises", "hrooms",
                "ifelectric", "iftelephone", "ifTV", "ifdiploma", "hhsize", "age",
                "localdist", "liv_for_6mth", "elecbill", "internet_type", "rent",
                "main_job")

# Transform categorical variables into binary indicators
df1$liv_for_6mth <- ifelse(df1$liv_for_6mth ==1 ,1 ,0)
df1$rent[is.na(df1$rent)] <- 0
df1$rent <- df1$rent/1000
df1$elecbill[is.na(df1$elecbill)] <- 0
df1$elecbill <- df1$elecbill/1000
df1$female <- ifelse(df1$female == 2 ,1 ,0)
df1$ifphone <- ifelse(df1$ifphone == 1, 1 ,0)
df1$ifinternet <- df1$inter_phone+df1$inter_office+df1$inter_cafe+df1$inter_home+df1$inter_scho
df1$ifinternet <- ifelse(df1$ifinternet >= 1,1,0)
df1$ifelectric <- ifelse(df1$ifelectric <= 3, 1, 0)
df1$iftelephone <- ifelse(df1$iftelephone == 1, 1, 0)
df1$ifTV <- ifelse(df1$ifTV == 1, 1, 0)

# Composite digital connectivity variable
df1$dig_connetivity <- df1$ifphone+df1$ifinternet+df1$ifTV+df1$iftelephone
df1$accessinfo <- ifelse(df1$dig_connetivity >= 1,1,0)

# Employment indicators
df1$work_self <- ifelse(df1$work_self == 1, 1, 0)
df1$work_gov <- ifelse(df1$work_gov == 1, 1, 0)
df1$work_trainee <- ifelse(df1$work_trainee == 1, 1, 0)
```

```
df1$inlf <- df1$work_self + df1$work_gov + df1$work_trainee
df1$job_status <- ifelse(df1$inlf >= 1, 1, 0)

# Other control variables
df1$ifrural <- ifelse(df1$ifrural == 2, 1 ,0)
df1$educ <- ifelse(df1$educ >= 3, 1, 0)
df1$ifmarried <- ifelse(df1$ifmarried == 2|df1$ifmarried ==3, 1, 0)
df1$ifdiploma <- ifelse(df1$ifdiploma > 0 ,1 ,0)
df1$main_job <- ifelse(df1$main_job == 1, 0, 1)
```

Dataset Preparation and Descriptive Statistics

```
# Create regression datasets
df1_reg <- df1[,c("female", "accessinfo", "job_status", "min_month_cost",
                  "ifrural", "age", "ifroad", "train_cen", "ifbus",
                  "ifmarried", "ifdiploma", "hhsize", "ifTV", "ifinternet",
                  "ifphone", "iftelephone", "localdist","liv_for_6mth",
                  "elecbill", "rent", "Enterprises", "phone1", "phone2", "phone3")]

df2_reg <- df1[,c("female", "main_job", "min_month_cost", "ifrural", "age",
                 "ifroad", "train_cen", "ifbus", "ifmarried", "ifdiploma",
                 "hhsize", "ifTV", "ifinternet","ifphone", "iftelephone",
                 "localdist","liv_for_6mth","elecbill", "rent",
                 "Enterprises","accessinfo")]

# Remove missing values
df1_reg <- na.omit(df1_reg)
df2_reg <- na.omit(df2_reg)

# Create additional variables
df1_reg$lnminmonthcost <- log(df1_reg$min_month_cost)
df1_reg$age2 <- df1_reg$age^2

# Generate descriptive statistics
df1_reg %>% stargazer(type = "html", out = "de_stat.html")
df2_reg %>% stargazer(type = "html", out = "de_stat2.html")
```

Linear Probability Models

```
# Simple linear probability model
```

```
simple_linear <- lm(job_status ~ accessinfo, data = df1_reg)
summary(simple_linear)
print(round(coeftest(simple_linear, vcov=vcovHC(simple_linear)),4))

# Multiple linear probability model
multiple_linear <- lm(job_status ~ accessinfo + ifrural + ifroad + localdist
                      + liv_for_6mth + train_cen + ifbus + ifmarried + female
                      + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                      + Enterprises, data = df1_reg)
summary(multiple_linear)
print(round(coeftest(multiple_linear, vcov=vcovHC(multiple_linear)),4))

# Calculate robust standard errors
robust_se_simple <- sqrt(diag(vcovHC(simple_linear, type = "HC1")))
robust_se_multiple <- sqrt(diag(vcovHC(multiple_linear, type = "HC1")))

# Generate regression table
stargazer(simple_linear, multiple_linear,
          type = "html",
          se = list(robust_se_simple, robust_se_multiple),
          title = "Simple and Multiple LPM with Robust SEs",
          align = TRUE,
          out = "Single&Multiple.html")
```

Instrumental Variables Analysis

```
# First stage regression
test.regression <- lm(accessinfo ~ elecbill + phone1 + phone2 + phone3
                      + ifrural + ifroad + localdist + liv_for_6mth + train_cen
                      + ifbus + ifmarried + female + age + age2 + lnminmonthcost
                      + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)
print(round(coeftest(test.regression, vcov=vcovHC(test.regression)),4))

# Test instrument relevance
linearHypothesis(test.regression, c("elecbill=0","phone1=0","phone2=0","phone3=0"))

# IV regression with 4 instruments
ivreg <- ivreg(job_status ~ accessinfo + ifrural + ifroad + localdist
               + liv_for_6mth + train_cen + ifbus + ifmarried + female
               + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
```

```
               + Enterprises|elecbill + phone1+ phone2 + phone3
               + ifrural + ifroad + localdist + liv_for_6mth + train_cen
               + ifbus + ifmarried + female + age + age2 + lnminmonthcost
               + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)
print(round(coeftest(ivreg, vcov=vcovHC(ivreg)),4))
summary(ivreg, vcov = vcovHC(ivreg), diagnostics = TRUE)

robust_se_IV <- sqrt(diag(vcovHC(ivreg, type = "HC1")))
```

Overidentification Tests

```
# Overidentification tests
df1_reg$resid_iv <- ivreg$residuals
overid1 <- lm(resid_iv ~ elecbill + phone1 + phone2 + phone3
             + ifrural + ifroad + localdist + liv_for_6mth + train_cen
             + ifbus + ifmarried + female + age + age2 + lnminmonthcost
             + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)

# Function to extract test results
extract_test_results <- function(test_result, hypothesis_name) {
  chisq_stat <- test_result$Chisq[2]
  p_value <- test_result$`Pr(>Chisq)`[2]

  return(data.frame(
    Hypothesis = hypothesis_name,
    Chi_Square = round(chisq_stat, 4),
    P_Value = round(p_value, 4),
    stringsAsFactors = FALSE
  ))
}

# Test various instrument combinations
test1 <- linearHypothesis(overid1, c("elecbill=0","phone1=0","phone2=0","phone3=0"),
                          test = "Chisq", vcov = vcovHC(overid1))
test6 <- linearHypothesis(overid1, c("elecbill=0","phone1=0"),
                          test = "Chisq", vcov = vcovHC(overid1))
# [Additional tests omitted for brevity]

results_list <- list(
  extract_test_results(test1, "elecbill=0, phone1=0, phone2=0, phone3=0"),
```

```
  extract_test_results(test6, "elecbill=0, phone1=0")
  # [Additional results omitted for brevity]
)

overid_results <- do.call(rbind, results_list)
print(overid_results)
```

Refined IV Model and Hausman Test

```
# IV regression with 2 instruments (based on overid tests)
ivregnew <- ivreg(job_status ~ accessinfo + ifrural + ifroad + localdist
                  + liv_for_6mth + train_cen + ifbus + ifmarried + female
                  + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                  + Enterprises|elecbill + phone1
                  + ifrural + ifroad + localdist + liv_for_6mth + train_cen
                  + ifbus + ifmarried + female + age + age2 + lnminmonthcost
                  + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)
summary(ivregnew, vcov = vcovHC(ivregnew),diagnostics = TRUE)

# Hausman endogeneity test
testnew <- lm(accessinfo ~ elecbill + phone1
              + ifrural + ifroad + localdist + liv_for_6mth + train_cen
              + ifbus + ifmarried + female + age + age2 + lnminmonthcost
              + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)

Hausement <- lm(job_status ~ accessinfo + testnew$residuals +ifrural + ifroad + localdist
                + liv_for_6mth + train_cen + ifbus + ifmarried + female
                + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                + Enterprises, data = df1_reg)
summary(Hausement, vcov = vcovHC(Hausement), diagnostic = TRUE)

linearHypothesis(testnew, c("elecbill=0","phone1=0"))
```

Component-Level Analysis

```
# Decompose information access into individual components
decomp1 <- lm(job_status ~ ifphone + ifrural + ifroad + localdist
              + liv_for_6mth + train_cen + ifbus + ifmarried + female
              + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
              + Enterprises, data = df1_reg)
```

```
decomp1.iv <- ivreg(job_status ~ ifphone + ifrural + ifroad + localdist
                    + liv_for_6mth + train_cen + ifbus + ifmarried + female
                    + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                    + Enterprises|phone1 + elecbill
                    + ifrural + ifroad + localdist + liv_for_6mth + train_cen
                    + ifbus + ifmarried + female + age + age2 + lnminmonthcost
                    + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)

# Similar analysis for internet, TV, and telephone
# [Additional component analyses omitted for brevity]

# Function to create systematic robust standard errors
create_se_vector <- function(model) {
  coef_names <- names(coef(model))
  robust_test <- coeftest(model, vcov = vcovHC(model, type = "HC1"))
  se_vector <- robust_test[, "Std. Error"]
  names(se_vector) <- rownames(robust_test)
  se_ordered <- se_vector[coef_names]
  return(se_ordered)
}

# Extract IV diagnostics
extract_iv_diagnostics <- function(iv_model, model_name) {
  iv_summary <- summary(iv_model, vcov = vcovHC(iv_model), diagnostics = TRUE)
  diagnostics <- iv_summary$diagnostics

  return(list(
    model = model_name,
    weak_stat = diagnostics[1, "statistic"],
    weak_pval = diagnostics[1, "p-value"],
    wu_stat = diagnostics[2, "statistic"],
    wu_pval = diagnostics[2, "p-value"],
    sargan_stat = diagnostics[3, "statistic"],
    sargan_pval = diagnostics[3, "p-value"],
    observations = nobs(iv_model)
  ))
}
```

Alternative Variable Construction and Robustness Checks

```
# Alternative construction of information access variable
df1_reg$accessinfo_new <- df1_reg$ifphone + df1_reg$ifTV + df1_reg$iftelephone
df1_reg$accessinfo_new <- ifelse(df1_reg$accessinfo_new >= 1,1,0)

new_model <- lm(job_status ~ accessinfo_new + ifrural + ifroad + localdist
                + liv_for_6mth + train_cen + ifbus + ifmarried + female
                + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                + Enterprises, data = df1_reg)

ivreg_new <- ivreg(job_status ~ accessinfo_new + ifrural + ifroad + localdist
                    + liv_for_6mth + train_cen + ifbus + ifmarried + female
                    + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                    + Enterprises | elecbill + phone1
                    + ifrural + ifroad + localdist + liv_for_6mth + train_cen
                    + ifbus + ifmarried + female + age + age2 + lnminmonthcost
                    + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)

# Alternative dependent variable analysis
df2_reg$accessinfo_new <- df2_reg$ifphone + df2_reg$ifTV + df2_reg$iftelephone
df2_reg$accessinfo_new <- ifelse(df2_reg$accessinfo_new >= 1, 1, 0)
df2_reg$age2 <- df2_reg$age^2
df2_reg$lnminmonthcost <- log(df2_reg$min_month_cost)

mainjob <- lm(main_job ~ accessinfo_new + ifrural + ifroad + localdist
              + liv_for_6mth + train_cen + ifbus + ifmarried + female
              + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
              + Enterprises, data = df2_reg)
```

Subgroup Analysis

```
# Subgroup analysis by household size and monthly cost
small.hhsize <- subset(df1_reg, hhsize < 10)
large.hhsize <- subset(df1_reg, hhsize >= 10)
low.cost <- subset(df1_reg, min_month_cost < 110000)
high.cost <- subset(df1_reg, min_month_cost >= 110000)

# Run regressions for each subgroup
small.hhsize <- lm(job_status ~ accessinfo_new + ifrural + ifroad + localdist
```

```
                    + liv_for_6mth + train_cen + ifbus + ifmarried + female
                    + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                    + Enterprises, data = small.hhsize)

large.hhsize <- lm(job_status ~ accessinfo_new + ifrural + ifroad + localdist
                    + liv_for_6mth + train_cen + ifbus + ifmarried + female
                    + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
                    + Enterprises, data = large.hhsize)

# Z-tests for subgroup differences
beta1_small <- coef(summary(small.hhsize, vcov=vcovHC(small.hhsize, type = "HC1")))["accessinfo
se1_small <- coef(summary(small.hhsize, vcov=vcovHC(small.hhsize, type = "HC1")))["accessinfo_n

beta1_large <- coef(summary(large.hhsize, vcov=vcovHC(large.hhsize, type = "HC1")))["accessinfo
se1_large <- coef(summary(large.hhsize, vcov=vcovHC(large.hhsize, type = "HC1")))["accessinfo_n

z_hhsize <- (beta1_small - beta1_large)/sqrt(se1_small^2 + se1_large^2)
p_hhsize <- 1 - pnorm(z_hhsize)
cat("Z =", round(z_hhsize, 3), "one_side_pvalue", round(p_hhsize, 4), "\n")
```

Logit Model and Control Function Approach

```
# Logit model
logit_new <- glm(formula = job_status ~ accessinfo_new + ifrural + ifroad + localdist
             + liv_for_6mth + train_cen + ifbus + ifmarried + female
             + age + age2 + lnminmonthcost + ifdiploma + rent + hhsize
             + Enterprises, family = binomial(link = "logit"), data = df1_reg)
summary(logit_new)

# Calculate average marginal effects
logitnew <- margins(logit_new)
summary(logitnew)

# Control function approach
cfa_1 <- lm(formula = accessinfo_new ~ elecbill + phone1
            + ifrural + ifroad + localdist + liv_for_6mth + train_cen
            + ifbus + ifmarried + female + age + age2 + lnminmonthcost
            + ifdiploma + rent + hhsize + Enterprises, data = df1_reg)
summary(cfa_1, vcov = vcovHC(cfa_1, type = "HC1"))
```

```
# Test instruments in first stage
linearHypothesis(cfa_1, c("elecbill=0", "phone1=0"), vcov = vcovHC(cfa_1, type = "HC1"))

# Add residuals from first stage
df1_reg$residual_hat <- cfa_1$residuals

# Second stage with control function
cfa_2 <- glm(formula = job_status ~ accessinfo_new + residual_hat + ifrural
             + ifroad + localdist + liv_for_6mth + train_cen + ifbus
             + ifmarried + female + age + age2 + lnminmonthcost + ifdiploma
             + rent + hhsize + Enterprises, family = binomial(link = "logit"), data = df1_reg)
summary(cfa_2)

# Calculate marginal effects for control function model
cfa2_margin <- margins(cfa_2)
summary(cfa2_margin)
```